FREDERIC OKE-AGBO, MS, MPH, ASA

Actuary - L&H Mortality, Biometric & Longevity Risks R&D | Experience Analysis

Associate of the Society of Actuaries (ASA) with 5+ years of experience specializing in mortality experience studies, assumption development, and biometric risk modeling within Life & Health reinsurance.Proven track record of translating complex experience data into actionable insights for pricing, valuation, and in-force portfolio management across North America life insurance markets. expertise in blending classical actuarial methodologies with advanced predictive modeling and public health epidemiological perspectives to model mortality, excess lapse, and policyholder behavior. Adept at championing cross-functional research initiatives, automating data pipelines and presenting key findings to global stakeholders.

RELEVANT EXPERIENCE

Nov 2021 | Present

Biometric Risks Modeling, Experience Analysis

SCOR GLOBAL Re — Charlotte, NC, USA

- Collaborated with a multidisciplinary global team to develop mortality trend assumptions for both longevity and protection businesses, incorporating a holistic view of future Morbidity drivers for pricing and valuation of in-force blocks.
- Led the estimation of Economic Capital for Post-Level Term (PLT) life products, focusing on excess lapse and mortality risk through predictive modeling.
- Built a model to quantify mortality trends resulting from delayed cancer screenings due to COVID-19 impacts on the U.S. healthcare system.
- Developed a predictive model for mortality trend assessment using risk embedded risk factors approach, supporting holistic forward-looking assumptions setting.
- Directed the implementation of lapse and mortality improvement assumptions, ensuring accurate integration into valuation, and pricing models.
- Accelerated the assumptions validation process by automating data preprocessing and model calibration using open-source R Shiny tools.
- Collaborated with global data analytics team to develop unified analytics products use for refining assumptions for pricing, assessing emerging trends in experience analysis and assessing the performance of treaties for in-force management.
- Led the development of policyholder's behaviors assumptions (lapse and future conversion) with integration of predictive modeling, resulting in a 15% improvement in assumption accuracy.
- Enhanced mortality risk analysis by integrating publicly available data via APIs, enriching data dimensions and insights.
- Applied advanced machine learning techniques to quantify the significance of new data points in mortality risk modeling.

CONTACT INFO

@fokeagbo

@fokeagbo

fredook15@gmail.com

Atlanta, GA, USA

SKILLS

- Mortality & Morbidity Research: Expertise in biometric risk modeling, mortality trend analysis, & morbidity insights.
- Experience Studies & Assumption: Skilled in conducting experience studies and setting robust, data-driven actuarial assumptions.
- Cross-Functional Collaboration: Work effectively with multidisciplinary global teams to drive projects to completion.
- ML & Predictive Analytics: Experience applying ML techniques for risk investigation, quantification, and scoring.
- Modeling: Time Series, GLM, GAM, Ensemble (Boosting, Random Forest), Decision Trees, PCA, Survival Models.
- Tools: Azure DevOps, Azure Databricks, R, SQL, and Python.
- Data & Visualization: Robust data cleaning, preprocessing, and visualization.
- Languages: English, French (Fluent).

Sept 2020 | Nov 2021

Actuarial Associate

American Equity, Investment and Life Insurance — Des Moines, IA, USA

- Designed SQL-based data pipelines to support actuarial modeling and improve data accuracy and accessibility.
- Developed experience-based assumption models for policyholder behavior, supporting model validation and pricing decisions.
- Built predictive models for partial withdrawals and life income rider utilization using statistical and machine learning techniques in R.
- Conducted model testing and validation to ensure reliability and alignment with actuarial standards.

Aug 2018 | Sept 2020

Graduate Teaching Assistant

University of Iowa — Iowa, USA

- Prepared lessons, summarized key learning points, provided academic support, conveyed complex concepts.
- Taught students how to apply statistical concepts to solve real world business problems.

EDUCATION

2020

University of Iowa

M.S., Actuarial Science — Iowa, USA

2015

University Libre de Bruxelles

M.P.H. (Dist), Public Health Methodologies — Brussels, Belgium

VOLUNTEERING LIFE

Dec 2015 | Present

Member of Medical Reserve Corps

DeKalb Department of Public Health — Georgia, USA

Participated in training on incident preparedness and management, to support the local community during times of disasters

AWARDS

Apr 2020

Robert H Taylor-Knowler Award

Outstanding achievement in Life Contingencies.